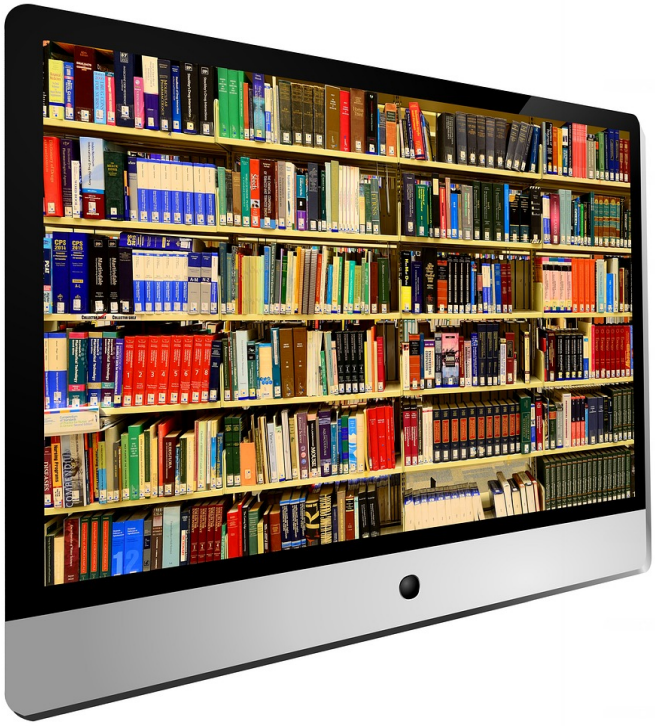
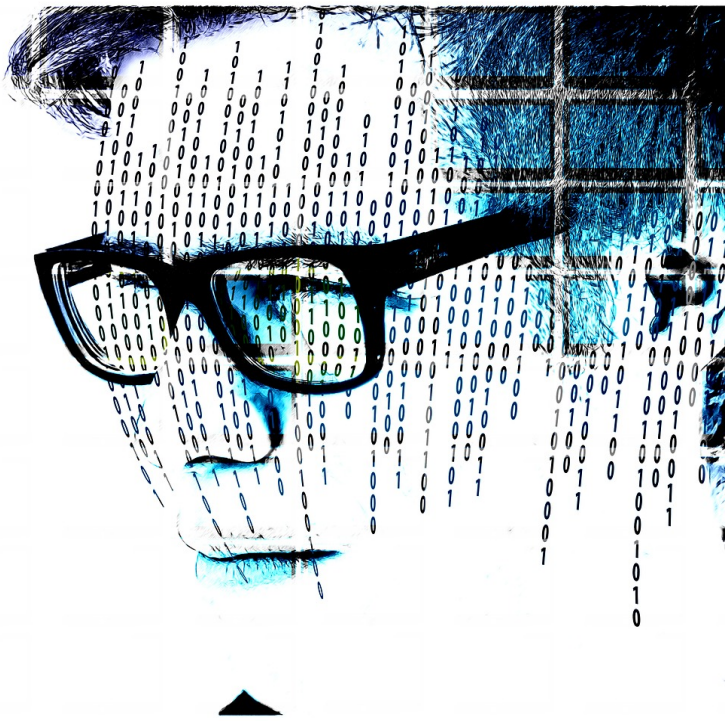
THE VIRTUAL MEET EXPERIENCE

2019 - 2020

HS VIRTUAL CHALLENGE MEET #3





COMPUTER SCIENCE

DO NOT OPEN TEST UNTIL TOLD TO DO SO

2019-2020 Virtual Challenge Meet #3 – Computer Science

Note: Correct responses are based on Java, **J2sdk v 1.7.25**, from Sun Microsystems, Inc. All provided code segments are intended to be syntactically correct, unless otherwise stated (i. e. `error` is an answer choice) and any necessary Java 2 Standard Packages have been imported. Ignore any typographical errors and assume any undefined variables are defined as used. **For all output statements, assume that the System class has been statically imported…** ***import static java.lang.System.*;***

QUESTION 1

Which of the following is equivalent to the decimal value `17`?

A. 10001_2 B. 101_4 C. 21_8 D. 11_{16} E. All are equivalent

QUESTION 2

What is output by the code to the right?

A. `2` B. `4` C. `6`

D. There is no output due to a compile error.

E. There is no output due to a runtime error.

```
out.println(12 / 3+2);
```

QUESTION 3

What is output by the code to the right?

A. `21|1d|111`

B. `100|1d|111`

C. `21|1d|false`

D. `21|1D|true`

E. There is no output due to an error.

```
out.printf("%o|%X|%b",17,29,7);
```

QUESTION 4

What is output by the code to the right?

A. `M` B. `I` C. `U` D. `T`

E. There is no output due to a compile error.

```
String s="IMMUTABLE";
s.substring(3);
out.println(s.charAt(1));
```

QUESTION 5

What is output by the code to the right?

A. `True` B. `False` C. `false` D. `true`

E. There is no output due to a syntax error.

```
boolean a=true;
boolean b=!!!a;
boolean c=b && b||a;
out.println(c);
```

QUESTION 6

What is output by the code to the right?

A. `6.0` B. `5` C. `5.0` D. `6`

E. There is no output due to an error.

```
int y=5;
double g=6;
out.println(Math.min(y, g));
```

QUESTION 7

What is the output by the code to the right?

A. `23` B. `12`

C. `1` D. `13`

E. `3`

```
int y=8;
if(y%2==0)out.print(1);
else  out.print(2);
      out.print(3);
```

QUESTION 8

What is the output by the code to the right?

A. 36
B. 38
C. 33
D. 35
E. 39

```
int n=2;
for(int y=0;y<12;y+=1)
  n+=3;
out.println(n);
```

QUESTION 9

What is output by the code to the right?

A. Integer.MAX_VALUE
B. 0
C. Integer.MIN_VALUE
D. 1
E. There is no output due to an error.

```
String s="";
for(int g=0;g<0;g--)
  s+="*";
out.println(s.length());
```

QUESTION 10

What is the output by the code to the right ?

A. [1, 6, 9, 16, 5]
B. [1, 2, 4, 7, 5]
C. [2, 11, 9, 16, 5]
D. [2, 6, 9, 7, 5]
E. There is no output due to an error.

```
int[] i= {1, 2, 4, 7, 5};
i[1]+=i[2]+=i[4];
i[0]++;
i[3]+=i[2];
out.println(Arrays.toString(i));
```

QUESTION 11

What is the output by the code to the right?

A. 1
B. 5
C. 0
D. There is no output due to a compile error.
E. There is no output due to a runtime error.

```
String s="12q34e56t78u90";
Scanner sc=new Scanner(s);
sc.useDelimiter("[a-z]");
int y=0;
while(sc.hasNext()) {
  sc.next();
  y++;
}
out.println(y);
```

QUESTION 12

What is the output by the code to the right ?

A. 23
B. 28
C. 40
D. 44
E. There is no output due to an error.

```
out.println(9 + 7 | 4 * 6 ^ 4);
```

QUESTION 13

What is the correct order of precedence for the operators to the right ?

A. II, IV, I, III
B. IV, I, II, III
C. I, IV, II, III
D. III, II, I, IV
E. II, I, IV, III

```
I. ++(post)
II. new(object creation)
III. <=
IV. ++(pre)
```

QUESTION 14

What is the output by the code to the right ?

A. 4
B. 16
C. 8
D. 2
E. 32

```
System.out.println(Short.SIZE);
```

QUESTION 15

What is the output by the code to the right ?

A. `you do for me`

B. `What it does`

C. `What it do`

D. `I appreciate what`

E. `Hello There`

```
ArrayList<String>a;
a=new ArrayList<String>();
a.add("Hello There");
a.add("What it do");
a.add("What it does");
a.add("Cancelled");
a.add("I appreciate what");
a.add("you do for me");
Collections.sort(a);
out.println(a.get(3));
```

QUESTION 16

What is the output by the code to the right ?

A. `true true` B. `false true`

C. `true false` D. `false false`

E. There is no output due to a syntax error.

```
Integer a=212;
Integer b=a;
out.print((a==b)+" ");
b=212;
out.println(a==b);
```

QUESTION 17

What is the output by the code to the right ?

A. `25` B. `2345`

C. `245` D. `5`

E. There is no output due to an error.

```
long g=45;
switch(g) {
case 34:out.print(1);
case 45:out.print(2);
case 46:out.print(3);
default:out.print(4);
}
out.print(5);
```

QUESTION 18

What is output by the code to the right ?

A. `12` B. `10`

C. `5` D. `6`

E. There is no output due to an error.

```
ArrayList<Integer>a;
a=new ArrayList<Integer>();
for(int y=1;y<9;y++)a.add(y);
a.replaceAll(n -> n*2);
out.println(a.get(5));
```

QUESTION 19

What is output by the code to the right?

A. `Hello World` B. `Hello`

C. `Hello W o r l d` D. No output due to an error.

E. Output cannot be determined until runtime.

```
String f="Hello";
char[] c= {'W','o','r','l','d'};
out.println(f+" "+c);
```

QUESTION 20

What is the value of `0b10010`?

A. `17` B. `18` C. `9` D. `10` E. None of the above.

QUESTION 21

What is the output by the code to the right ?

A. `il122 1221R22s1022hawn1221221222221L12`
B. `hawn2122L122il122 1221R22s102`
C. `il222 2221R22s1022hawn1211221222121L12`
D. `hawn2122L222il222 1111R22s102`
E. There is no output due to a runtime error.

```
String k="Lil R0shawn";
int len=k.length();
for(int y=0;y<3;y+=(1+y)*3/2) {
  for(int j=0;j<3;j++) {
     int min=Math.min(j%len, y%len);
     int max=Math.max(j%len, y%len);
     k=k.substring(max)+"2"
        +k.substring(min,max)+"1"
        +k.substring(0,min)+"2";
  }
}
out.println(k);
```

QUESTION 22

Which of the following could replace **<1*>** in the code to the right to make the affected methods accessible in all classes?

A. `abstract`
B. `super`
C. `public`
D. `interface`
E. `protected`

QUESTION 23

Which of the following could replace **<2*>** in the code to the right?

A. `super()`
B. `super(n)`
C. `super(y, n)`
D. Nothing is needed.
E. More than one of the above.

QUESTION 24

Assuming that **<1*>** and **<2*>** are filled correctly, what is output by `//1` ?

A. `Charlie: 5 :: 7`
B. `Bonny: 5`
C. `Charlie: 5 :: 5`
D. `Charlie: 7 :: 7`
E. This is no output due to an error.

QUESTION 25

Assuming that **<1*>** and **<2*>** are filled correctly, what is output by `//2` ?

A. `Charlie`
B. `5`
C. `7`
D. Output cannot be determined until runtime
E. This is no output due to a runtime error.

```
class Parent{
  private int y=5;
  String name;
  <1*> Parent(String n) {
     name=n;
  }
  <1*> int getY() {
     return y;
  }
  <1*> String toString() {
     return name+": "+y;
  }
}
class Child extends Parent{
  private int y;
  <1*>Child(int h,String n) {
     <2*>;
     y=h;
  }
  <1*> String toString() {
     return super.toString()
     +" :: "+y;
  }
}
/////////////client code/////////////
Parent p=new Parent("Bonny");
Child c=new Child(7,"Charlie");
out.println(c);    //1
out.println(c.getY());    //2
```

QUESTION 26

What is output by the code to the right?

A. `false true false`
B. `true true true`
C. `true false true`
D. `false true true`
E. `false false true`

```
String a="Captain Daniel";
String b="Captain Dayniel";
String c="Captain Dwayne";
String s1="[A-D][a-z]+ [A-D][a-z]*";
String s2=".{14}";
String s3="Captain Dwayne";
out.print(a.matches(s1)+" ");
out.print(b.matches(s2)+" ");
out.println(c.matches(s3));
```

QUESTION 27

What is the Big O runtime of the code to the right?

A. O(n)
B. O(nlogn)
C. O(logn)
D. O(n^2)
E. None of the above.

```
int n=7;
for(int m=0;m<n;m++)
  for(int v=0;v<m;v+=2)
          out.print(0);
```

QUESTION 28

What is output by the line marked //q28?

A. 15
B. 6
C. 7
D. 1
E. 14

QUESTION 29

What is output by the line marked //q29?

A. 1
B. 0
C. 3
D. -1
E. 2

QUESTION 30

What is output by the line marked //q30?

A. 128
B. 256
C. 63
D. 255
E. 127

```
int re(int y) {
  if(y<0)
     return 1;
  else
     return 1+2*re(y-2);
}
////////////client code///////////////
out.println(re(4)); //q28
out.println(re(-1)); //q29
out.println(re(10)); //q30
```

QUESTION 31

Which of the following can replace **<1*>** in the code to the right?

A. 5.0
B. 5
C. (float)5
D. 5.0f
E. All of the above.

```
Double d=<1*>;
```

QUESTION 32

What is output by the line marked //q32?

A. 12
B. 10
C. 2
D. 3
E. 5

QUESTION 33

What is output by the line marked //q33?

A. [3, 7, 5, 9, 10, 12]
B. [3, 5, 7, 9, 10, 12]
C. [2, 3, 7, 5, 9, 10]
D. [2, 7, 5, 9, 10, 12]
E. [2, 3, 5, 7, 9, 10]

```
PriorityQueue<Integer>p;
p=new PriorityQueue<>();
p.add(5);
p.add(7);
p.add(2);
p.add(3);
p.add(10);
p.add(12);
p.add(9);
out.println(p.poll());    //q32
out.println(p);    //q33
```

QUESTION 34

What is the post order traversal of the binary tree to the right?

A. ABCDEFGHIJ
B. ABDGEHICFJ
C. GDHIEBJFCA
D. GDBHEIACFJ
E. LIHGFEDCBA

```
          A
         / \
        B   C
       / \   \
      D   E   F
     /   / \   \
    G   H   I   J
```

QUESTION 35

What is outptut by the code to the right ?

A. 269
B. 267
C. 144
D. 142
E. 162

```
int sum=2;
for(int y=0;y<7;y++) {
 for(int x=y;x<y*3;x+=3) {
  for(int h=0;h<10;h++)
    sum+=2;
  sum-=2;
 }
 sum-=3;
}
out.println(sum);
```

QUESTION 36

Which of the following matches the truth table to the right ?

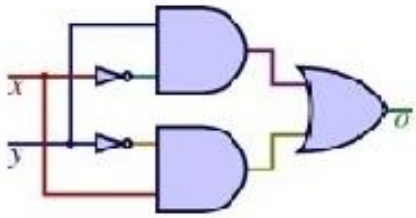
A. B || A && C
B. A || B && C
C. C || B && A
D. B && A && C
E. B || A || C

A	B	C	D
1	1	1	1
1	1	0	1
1	0	1	1
1	0	0	1
0	1	1	1
0	1	0	0
0	0	1	0
0	0	0	0

QUESTION 37

Which of the following is equivalent to the digital electronics diagram to the right?

A. x && !y
B. (x || !y) && (!x || y)
C. x && !y || !x && !y
D. x || !y && !x || y
E. x && !y || !x && y



QUESTION 38

What is the value of the post order expression to the right?

A. -7
B. 421
C. 532
D. 17
E. -328

5 6 + 7 - 8 9 * + 1 6 + *

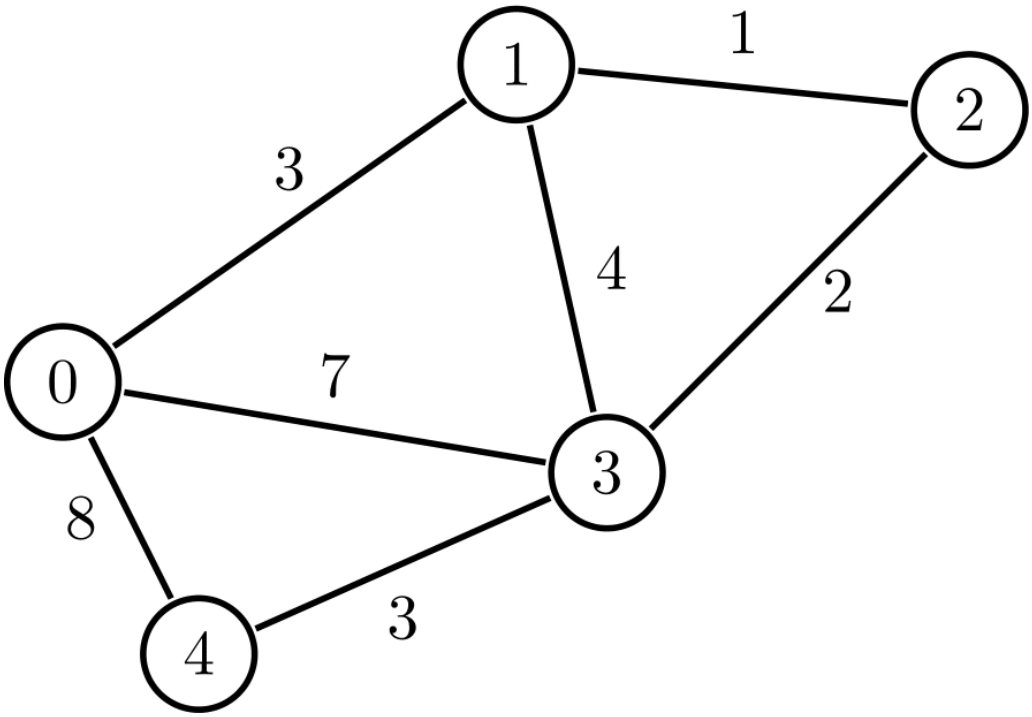
QUESTION 39

What is the fully simplified form of the following boolean expression ?

```
A || (B && (C ^ !B) && (A && B) && !B)
```

QUESTION 40

What is the shortest cost path from node 0 to 2 in the graph below ?



Name _____ Final Score _____ _____

School _____ 2nd Grading _____ _____

Division: 1A 2A 3A 4A 5A 6A Grade: 9 10 11 12 1st Grading _____ _____
score initials

The Virtual Challenge Meets

Computer Science - Student Answer Sheet

1) _____
2) _____
3) _____
4) _____
5) _____
6) _____
7) _____
8) _____
9) _____
10) _____

11) _____
12) _____
13) _____
14) _____
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27) _____
28) _____
29) _____
30) _____

31) _____
32) _____
33) _____
34) _____
35) _____
36) _____
37) _____
38) _____
39) _____
40) _____

Total Number Correct _____ x 6 = _____

Total Number Incorrect _____ x -2 = _____

Final Score (record here and top right) _____

2019-2020 Virtual Challenge Meet #3
Computer Science - KEY

1) E
2) C
3) D
4) A
5) D
6) C
7) D
8) B
9) B
10) C

11) B
12) B
13) C
14) B
15) C
16) C
17) E
18) A
19) E
20) B

21) B
22) C
23) B
24) A
25) B
26) C
27) D
28) A
29) A
30) E

31) A
32) C
33) A
34) C
35) A
36) B
37) E
38) C
39) A
40) 4

Note to Graders:

- All provided code segments are intended to be syntactically correct, unless otherwise stated (e.g. error is an answer). **Ignore any typographical errors.**
- Any necessary Standard Java 2 Packages are assumed to have been imported as needed.
- Assume any undefined (undeclared) variables have been defined as used.

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